Independent Audit of `phase2/good_twobody.tex`: Two-Soliton Conditional Ingredients, Counterterm Responses, Three-Body Clustering, and Near-Resonance Asymptotics

Independent Audit Report
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Abstract

We present an independent mathematical and numerical audit of `phase2/good_twobody.tex` (and `phase2/good_twobody.pdf`) together with its five verification scripts `phase2/scripts/good_twobody_*.py` and six cached JSON artefacts. All analytic derivations have been re-derived from scratch in `SymPy` and all numerical quantities have been recomputed at 70–90 decimal digits in `mpmath` by the standalone script `phase2/scripts/audit_good_twobody.py`, which imports no code from the audited package and verifies via SHA-256 digests that no existing repository file is modified. Of the thirteen claims and audit items examined, **nine are confirmed** and **four are qualified** (due to coordinate-frame dependence of the counterterm antisymmetry, script-coverage shortcuts in three package scripts, the distinction between scalar/Darboux clustering and the Toda Krein–Wiener–Hopf PDE, and the $O(h \ln h) + O(h)$ subleading Taylor structure near the head-on Y-resonance). **Zero claims are refuted**, and all five package scripts as well as `audit_good_twobody.py` print `ALL CHECKS PASSED`.

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1 Executive Summary and Verdict Table

Following `AGENTS.md`, every nontrivial claim is labelled:

- **(a)** proved in the literature;
- **(b)** conjectured or heuristic in the literature;
- **(c)** derived and verified in this project;
- **(d)** a guess or open hypothesis.

For each claim in `phase2/good_twobody.tex` (Table 1 and text), we assign an independent audit verdict: **Confirmed**, **Qualified**, or **Refuted**.

Claim / Item	Label	Verdict	Independent Audit Summary & Evidence
C1. Classical Darboux shifts & $\partial_{P_j}\delta_0 = -\Delta X'_j$ (Eqs. (3)–(10))	(c)	**Qualified** (math confirmed; script check numerical)	Confirmed symbolically across all four orientation branches $(\epsilon_1,\epsilon_2)\in\{\pm1\}^2$ via canonical exponential reduction in `audit_good_twobody.py`. *Qualification*: `good_twobody_counterterms.py:54-58` only tested $\partial_{P_j}\delta_0 = -\Delta X'_j$ numerically via `mp.diff` at $(0.3, 0.6)$ rather than symbolically in `SymPy`.
C2. Continuum integrals I_{12}, J_{12} & dilogarithm form (Eqs. (11)–(14))	(c)	**Confirmed**	Re-derived from the Hirota four-wave phonon shift $\theta(q)$ and verified by independent hyperbolic-sinh quadrature ($q=\sinh u$) against the $\chi_2(z)=\frac{1}{2}(\mathrm{Li}_2(z)-\mathrm{Li}_2(-z))$ formula across all four sign branches; max difference 4.81×10^{-72}.
C3. One-body charges $E_{\rm m}, p_{\rm m}, L_1, p_{\rm can}$ & recoil $R[p]$ (Eqs. (15)–(17))	(c)	**Qualified** (formulas confirmed; mode-trace shortcut)	Closed forms $E_{\rm m}=(4-3\sqrt{3}\sin\alpha)/(6\pi)$, $p_{\rm m}=-vE_{\rm m}$, and $p_{\rm can}=\partial_v L_1 = p_{\rm m} - v\sin\alpha(\frac{2}{3}-\frac{\sqrt{3}}{4}\sin\alpha)/\pi$ confirmed to $<10^{-68}$. *Qualification*: `good_twobody_kernels.py:136-141` checks $E_{\rm m}$ using an algebraically pre-subtracted integrand with $-3k/(2\pi)$ inserted by hand; `audit_good_twobody.py` supplies the missing cutoff-regulated trace check with B_Λ and $C_\Lambda M$.
C4. Counterterm responses $K_\lambda, K_\eta, K_\zeta, K_\kappa$ at fixed P_j (Eqs. (18)–(20), Point 7)	(c)	**Qualified** (values confirmed; frame & PDE-test qualification)	Re-derived directly from the deformed Hirota bilinear equation $(\frac{1}{a}D_t^2 - bD_x^2 + dD_x^4)\tau\cdot\tau = 0$ without coordinate rescaling. At $(\alpha_1,\alpha_2)=(0.3,0.6)$: $K_\lambda^{++}=+0.050081404868418910$ and $K_\lambda^{+-}=-0.225173938683341758$ (confirmed to 80 digits). *Qualifications*: (i) `good_twobody_counterterms.py:62-78` differentiates the already-scaled expression $s\,\delta_0(P_1/p_*, P_2/p_*)$ rather than deformed PDE profiles; (ii) pure antisymmetry $K_\lambda(1,2)+K_\lambda(2,1)=0$ relies on the chamber-reversing Darboux frame where the symmetric coboundary $S_{\rm cb}=\frac{1}{2}(\Pi_1 M_2+\Pi_2 M_1)$ was subtracted; in a fixed-chamber incoming frame, $K_\lambda^{\rm sym}=3S_{\rm cb}-4\sum_j P_j\partial_{P_j}S_{\rm cb}=+0.271191727118\ldots\neq0$ at $(0.3,0.6,++)$.
C5. Connected insertion $\int(u_{12}^2-u_1^2-u_2^2)\,dx\,dt = \frac{4}{3}(v_2-v_1)\ln A_{12} + 2T_{12}$ (Eq. (21))	(c)	**Confirmed**	Derived via the Bäcklund telescoping density $u=\partial_x(\varphi_+-\varphi_-)$ and verified by independent 2D Gauss–Legendre spacetime integration of the exact two-soliton profile $u_{12}(x,t)=2\partial_x^2\ln\tau_2(x,t)$; $T=3000$ error is 4.74×10^{-51} for $(++)$ and $<10^{-70}$ for $(+-)$.

Claim / Item	Label	Verdict	Independent Audit Summary & Evidence
C6. One-loop n-insertion Feynman ring UV finiteness for $n \geq 2$	(c)	**Qualified** (true; untested in package scripts)	Confirmed symbolically in `audit_good_twobody.py`: the n-insertion ring integrand has denominator $(\omega_{\mathrm{E}}^2 + q^2 + q^4)^n$, giving UV superficial degree $3 - 2n$ (convergent for $n \geq 2$), with exact ω_{E}-integral $-\frac{(2n-3)!!}{2n\,(2n-2)!!}\frac{(6U)^n q^{2n}}{(q^2+q^4)^{n-1/2}}$. *Qualification*: asserted in `good_twobody.tex:240-244` without a script check in `good_twobody_*.py`.
C7. Three-soliton Hirota 4-wave phonon transmission factorisation $T_{123} = T_1T_2T_3$ (Point 8)	(c)	**Confirmed**	All $3^4 = 81$ bilinear polynomial coefficients of $\mathcal{P}(\sum_{\alpha=1}^4 \sigma_\alpha K_\alpha, \sum_{\alpha=1}^4 \sigma_\alpha W_\alpha)$ in the four-wave Hirota ansatz (3 solitons + 1 linearized phonon) vanish identically in `SymPy`, confirming $A_{j4} = -P(K_j - \mathrm{i}q, W_j - \mathrm{i}\omega)/P(K_j + \mathrm{i}q, W_j + \mathrm{i}\omega)$ and $T_{123}(q) = T_1(q)T_2(q)T_3(q)$.
C8. Three-body clustering vs. antisymmetric pair phase F_{12} (Point 8)	(c)	**Qualified** (reconciled with Toda audit; line 94–95 tautology)	Scalar clustering $S_3 = S_{12}S_{13}S_{23}$ leaves any pair phase F_{ij} unfixed; Darboux 3×3 minor clustering ($\det_{3\times3}\Phi = 0$ in `audit_frame_dictionary.tex`) kills infinite-rank $A_\lambda(y_1, y_2)$ ($\det_{3\times3} \neq 0$) but *not* the rank-2 bilinear $F_{12} = c(P_1M_2 - P_2M_1)$ ($\det_{3\times3} = 0$). On the Toda side, the finite-density Fermi sea and linear dressing enforce the inhomogeneous Krein PDE $(\partial_1 + \partial_2)\widehat{\Phi}_q =$ rank 1, which excludes F_{12} ($\det_{2\times2} = 27\sqrt{10}/80 \neq 0$). Also, `good_twobody_threebody.py:94-95` only checks an addition tautology $(f_{12} + f_{13} + f_{23}) - (f_{23} + f_{13} + f_{12}) = 0$.
C9. Head-on Y-resonance $A_{12} \to \infty$ ($\alpha_1 + \alpha_2 = \pi/3 - h$, Eqs. (22)–(25), Point 10)	(c)	**Qualified** (limits confirmed; $O(h \ln h) + O(h)$ remainder)	Finite limits J_*, $B_{p_{\mathrm{m}}}$, B_λ and logarithmic slopes $C_{p_{\mathrm{m}}}, C_{p_{\mathrm{can}}}, C_\lambda$ confirmed to $< 10^{-53}$. *Qualification*: the remainder written as $o(1)$ in Eqs. (22)–(25) is $\sim 1.9 \times 10^{-11}$ at $h = 10^{-12}$ because of the exact subleading Taylor terms $-h[(\partial_{\alpha_2}C)\ln A_{12}(h) + \partial_{\alpha_2}B] = O(h\ln h) + O(h)$; subtracting both first-order terms reduces the $h = 10^{-12}$ residual to $< 3.1 \times 10^{-23}$.
C10. Overtaking limits $A_{12} \to 0^+$ & refutation of $\lvert\alpha_1 - \alpha_2\rvert = \pi/3$ (Point 10)	(c)	**Confirmed**	At the light equal-velocity threshold $\alpha_2 = \alpha_1 + h$ ($A_{12} \propto h^2 \to 0^+$), identical-particle cancellation yields $K_\lambda = -\frac{4}{3}\sin\alpha_1\, h \ln h + O(h) \to 0$. On the heavy Y-line $\alpha_1 + \alpha_2 = 2\pi/3 - h$ ($\alpha_2^* > \pi/3$), $v_2^* \neq v_1$ so $K_\lambda \sim \frac{2}{3}(v_2^* - v_1)\ln h \to +\infty$, and `quantum_phase` is rightly `null`. At $(\alpha_1, \alpha_2) = (0.2, 0.2 + \pi/3)$ overtaking, $A_{12} = 0.355465479849\ldots \neq 0, \infty$, confirming the refutation of the false $\lvert\alpha_1 - \alpha_2\rvert = \pi/3$ resonance.
C11. 270-row grid JSON & honest `null` entries (Point 9)	(c)/(d)	**Confirmed**	All 270 rows (180 overtaking + 90 head-on) have `physical_symmetric: null` and `physical_antisymmetric: null`; `good_twobody_threebody.json` has `full_threebody_determinant: null`; `good_twobody_resonance.json` has `quantum_phase: null` on the heavy branch. All 10 independent 80-digit spot checks across $(++, --, +-, -+)$ agree to $\leq 3.81 \times 10^{-54}$.
C12. Conventions & non-reliance on refuted results (Points 3, 5)	(a)/(c)	**Confirmed**	Strict adherence to the `AGENTS.md` good-equation convention, the Corrigan–Zambon dictionary ($u_{\mathrm{CZ}} = -3\varphi$, $L_{\mathrm{CZ}} = 9L$), and the defect s convention (scattering $s = \sigma_{\mathrm{vac}}$, spectrum $\eta\sigma_{\mathrm{vac}}$). Zero reliance on any claim refuted in `phase2/audit_*.tex` or `phase2/errata.tex`.
C13. Script execution & cache safety (Point 6)	(c)	**Confirmed**	All 5 package scripts and `audit_good_twobody.py` run cleanly in the foreground, print `ALL CHECKS PASSED`, and leave `git status` of all tracked files untouched.

2 Script Execution, Runtimes, and Non-Modification Check (Point 6)

Each of the five package scripts and our standalone audit script `phase2/scripts/audit_good_twobody.py` was executed individually in the foreground from the repository root (`python3 phase2/scripts/xxx.py`).

Before and after running `audit_good_twobody.py`, SHA-256 digests of all 13 audited files (`good_twobody.tex`, `good_twobody.pdf`, the 5 scripts, and the 6 JSON files) were computed and verified to be identical.

Script	Wall time (s)	Exit	Key diagnostic printed
`good_twobody_kernels.py`	36.00	0	`max independent quadrature error 5.9347298e-66`
`good_twobody_counterterms.py`	5.00	0	`max fixed-momentum finite-difference error 1.86`
`good_twobody_local_insertion.py`	9.00	0	`elapsed seconds 8.62`
`good_twobody_threebody.py`	11.00	0	`Exact four-wave bilinear coefficients checked:`
`good_twobody_grid.py`	71.75	0	`grid rows 270 max precision difference 8.300023`
`audit_good_twobody.py`	47.71	0	`all 13 audited file SHA-256 digests unchanged`

Table 2: Foreground execution runtimes and outputs of the five audited package scripts and the standalone audit script.

3 Conventions and Non-Reliance on Refuted Results (Points 3 and 5)

3.1 Good/Bad Dictionary, Corrigan–Zambon Rescaling, and Defect s Convention (Point 3)

In Check 1 of `audit_good_twobody.py`, we verified symbolically in `SymPy` that `good_twobody.tex` adheres throughout to the fixed `AGENTS.md` convention for the **good** Boussinesq equation:

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \varphi_x = 2(\ln\tau)_{xx}, \qquad \mathcal{L} = \frac{1}{2}\varphi_t^2 - \frac{1}{2}\varphi_x^2 + \varphi_x^3 - \frac{1}{2}\varphi_{xx}^2, \tag{1}$$

with canonical symplectic form $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\varphi$ ($\pi = \varphi_t$) defining the normalisation of $\hbar$. The one-soliton family with angle $\alpha \in (0, \pi/3)$ (the spectrally stable light branch where $V''(0) > 0$) and direction sign $\epsilon \in \{\pm 1\}$ has:

$$\begin{aligned} &k = \sin\alpha, \qquad v = \epsilon\cos\alpha, \qquad \ell = -\epsilon\cos 3\alpha = -v(1-4k^2), \qquad M = 2k, \qquad \Pi = -Mv, \\ &P = \frac{2}{3}vk^3 = \frac{2}{3}\epsilon\sin^3\alpha\cos\alpha, \qquad E = \frac{2}{15}k^3(5-4k^2), \qquad \frac{\partial P}{\partial\alpha} = \frac{2}{3}\epsilon\sin\alpha\sin 3\alpha, \end{aligned} \tag{2}$$

satisfying $\partial_\alpha E = v\,\partial_\alpha P$ and $\partial_k \Pi = -2\ell/M = -\ell/k$.

We also checked the three external convention dictionaries referenced in the repository:

1. **Good vs. bad analytic continuation**: Under $\alpha = \mathrm{i}\beta$, $\sin\alpha = \mathrm{i}\sinh\beta$, $\cos\alpha = \cosh\beta$, and $\cos 3\alpha = \cosh 3\beta$, the good dispersion polynomial $\mathcal{P}_{\mathrm{good}}(K, W) = W^2 - K^2 + K^4$ maps up to overall sign to the bad dispersion polynomial $\mathcal{P}_{\mathrm{bad}}(\kappa, \Omega) = \Omega^2 - \kappa^2 - \kappa^4$ with $k_{\mathrm{bad}} = \sinh\beta$, $v_{\mathrm{bad}} = \epsilon\cosh\beta$, $\ell_{\mathrm{bad}} = \epsilon\cosh 3\beta$.

2. **Corrigan–Zambon dictionary**: Under $u_{\mathrm{CZ}} = -3\varphi$, $\mathcal{L}_{\mathrm{CZ}} = \frac{1}{2}(u_{\mathrm{CZ},t}^2 - u_{\mathrm{CZ},x}^2 - \frac{2}{3}u_{\mathrm{CZ},x}^3 - u_{\mathrm{CZ},xx}^2) = 9\mathcal{L}$ identically.

3. **Defect s convention**: As established in `phase2/errata.tex` and `phase2/audit_phase2i.tex`, defect scattering matrices use the bare vacuum parameter $s = \sigma_{\mathrm{vac}}$, whereas the bound-state/spectral polynomial uses the sign-dressed combination $\eta\sigma_{\mathrm{vac}}$. The package `good_twobody.tex` does not conflate the two.

3.2 Non-Reliance on Refuted Claims from Earlier Audits (Point 5)

We cross-checked `good_twobody.tex` and `good_twobody_*.py` against all prior audit reports (`audit_phase2i.tex`, `audit_smx.tex`, `audit_det_projected.tex`, `audit_projected_measure.tex`,

audit_determinant_foundations.tex, audit_onebody_scheme.tex, audit_frame_dictionary.tex, and errata.tex). The package was specifically engineered to replace the refuted shortcuts of earlier exploratory notes:

- It does *not* assume the refuted claim that the counterterm $\Delta H = \lambda\hbar \int u^2 \, \mathrm{d}x$ vanishes on two-soliton scattering at fixed physical momenta (it proves $K_\lambda \neq 0$).
- It does *not* use the refuted overt resonance condition $|\alpha_1 - \alpha_2| = \pi/3$ (it explicitly refutes it in §7 and good_twobody_resonance.json).
- It does *not* set the uncomputed boundary-compatible two-body functional determinant or collective-radiation Schur complement $\det(A - BC^{-1}B^{\mathsf{T}})$ to zero; instead, it records physical_symmetric: null and physical_antisymmetric: null across all 270 grid rows and leaves the unknown antisymmetric boundary/projection phase B_{12} explicit in Eq. (17).
- It does *not* import any code or cached data from legacy phase1/ or phase2/ scripts.

4 Audit of Claims, Error Budgets, and Script Circularity (Points 1, 2, 4)

4.1 Error Budgets (Point 2)

The numerical error budgets in good_twobody_*.py are separated by source and are not artificially loosened:

- **Working precision vs. closed-form/quadrature**: good_twobody_kernels.py works at 80 decimal digits and compares the closed-form χ_2 formula for I_{12} against numerical quadrature to $< 10^{-62}$ (actual max error 5.93×10^{-66}; our independent hyperbolic quadrature $q = \sinh u$ agrees to 4.81×10^{-72}).
- **Dual-precision grid verification**: good_twobody_grid.py computes every one of the 270 grid rows twice, at workdps(55) and workdps(75), recording precision_recheck_max_abs_difference on every row (worst case 8.30×10^{-55}, asserted $< 10^{-48}$).
- **Finite-difference step error**: good_twobody_counterterms.py uses central differences at step $\varepsilon = 10^{-18}$ at 80 digits (truncation error $O(\varepsilon^2) \sim 10^{-36}$, measured max error 1.87×10^{-35}, asserted $< 10^{-30}$). In audit_good_twobody.py, we improved this with two-step Richardson extrapolation ($\varepsilon = 10^{-14}$ and $\varepsilon/2$), reducing the finite-difference error to 3.05×10^{-54}.
- **Spacetime domain truncation error**: good_twobody_local_insertion.py separates spatial quadrature from finite-time truncation $t \in [-T, T]$ by recording errors at $T = 800$ and $T = 1600$ (where tail corrections scale as $\mathrm{e}^{-2\min(k_1,k_2)|v_1-v_2|T}$), achieving 5.13×10^{-18} at $T = 800$ and 9.03×10^{-36} at $T = 1600$ for $(++)$. In audit_good_twobody.py, we used 2D Gauss–Legendre integration on $x \in [-110, 110]$ with $T = 2000$ and $T = 3000$, reaching 2.29×10^{-34} ($T = 2000$) and 4.74×10^{-51} ($T = 3000$) for $(++)$, and $< 10^{-70}$ for $(+-)$.

4.2 Circularity and Coverage Qualifications (Points 1 and 4)

While no formula in the package is false, our line-by-line inspection of the five package scripts uncovered **four script-coverage / near-circularity shortcuts** where a script tested a simplified surrogate rather than the full derivation claimed in good_twobody.tex. All four gaps have been closed by explicit symbolic or regulated checks in audit_good_twobody.py:

1. **Scale-ansatz shortcut in good_twobody_counterterms.py:62-78**: To verify Eq. (19) ($K_\lambda = 3\delta_0 + 4D$, etc.), the package script defines scaled_phase(P1, P2, a=1, b=1, d=1, g=1) by evaluating the undeformed function $\delta_0(P_1/p_*, P_2/p_*)$ and multiplying by $s = db^{3/2}/(\sqrt{a}g^2)$,

then finite-differentiates with respect to a, b, d, g. That only tests Euler's homogeneous-function chain rule on (s, p_*), assuming the scale dictionary (s, p_*) as an input! In Check 2a of `audit_good_twobody.py`, we instead solve the deformed Hirota bilinear dispersion $W_j^2/a - bK_j^2 + dK_j^4 = 0$ and one-soliton momentum integral $P_j = \frac{2b^{3/2}}{3\sqrt{a}g^2}\epsilon_j \sin^3\alpha_j \cos\alpha_j$ directly for the deformed wave numbers $K_j(P_j; a, b, d, g)$, frequencies $W_j(P_j; a, b, d, g)$, and Hirota two-soliton shift $\tilde{A}_{12} = -\tilde{\mathcal{P}}(K_1 - K_2, W_1 - W_2)/\tilde{\mathcal{P}}(K_1 + K_2, W_1 + W_2)$ without any coordinate rescaling. Differentiating the resulting deformed phase by Richardson extrapolation reproduces $(K_\lambda, K_\eta, K_\zeta, K_\kappa)$ to 3.05×10^{-54}.

2. **Numerical-only check of $\partial_{P_j}\delta_0 = -\Delta X_j'$ in `good_twobody_counterterms.py:54-58`:** The package checks the Darboux gradient identity $\partial_{P_j}\delta_0 = -\Delta X_j'$ only numerically at $(\alpha_1, \alpha_2) = (0.3, 0.6)$ using `mp.diff`. In Check 1b of `audit_good_twobody.py`, we proved symbolically in `SymPy` across all four orientation sectors $(\epsilon_1, \epsilon_2) \in \{\pm 1\}^2$ that

$$\frac{1}{\partial_{\alpha_1} P_1}\frac{\partial \delta_0}{\partial \alpha_1} + \Delta X_1' = 0, \qquad \frac{1}{\partial_{\alpha_2} P_2}\frac{\partial \delta_0}{\partial \alpha_2} + \Delta X_2' = 0 \tag{3}$$

hold as exact trigonometric identities, by proving that $\partial_{\alpha_1} \ln A_{12} = \frac{6 \sin\alpha_1 \sin 3\alpha_1 \cos 3\alpha_2}{\cos 3\alpha_1 - \epsilon_1\epsilon_2 \cos 3\alpha_2}$.

3. **Pre-subtracted integrand in `good_twobody_kernels.py:136-141`:** The package checks the one-body mode energy $E_{\rm m} = (4 - 3\sqrt{3}\sin\alpha)/(6\pi)$ against the integral $\int_0^\infty [\sqrt{1+q^2}\,\theta'(q)/(2\pi) - 3k/(2\pi)]\,\mathrm{d}q$ in which the counterterm subtraction $-3k/(2\pi)$ has already been inserted by hand. In Check 3b of `audit_good_twobody.py`, we evaluate the raw cutoff-regulated mode trace on $[0, \Lambda]$ at $\Lambda = 500$ and $\Lambda = 1000$,

$$E_{\rm reg}(\Lambda) = \int_0^\Lambda \frac{\mathrm{d}q}{2\pi}\sqrt{1+q^2}\,\theta'(q) + C_\Lambda M, \qquad C_\Lambda = -\frac{3\Lambda}{4\pi}, \tag{4}$$

confirming that B_Λ cancels when subtracting the vacuum, $C_\Lambda M = -\frac{3k}{2\pi}\Lambda$ cancels the linear divergence of the phase-shift trace, and the $O(\Lambda^{-2})$ Richardson combination $(4E_{\rm reg}(2\Lambda) - E_{\rm reg}(\Lambda))/3$ matches $E_{\rm m}$ to $< 10^{-17}$.

4. **Unchecked UV ring power-counting and addition tautology in `good_twobody_threebody.py:94-9`** The one-loop n-insertion ring UV finiteness claim in `good_twobody.tex:240-244` had no test in the package scripts (supplied symbolically in Check 2c of `audit_good_twobody.py`), and lines 94–95 of `good_twobody_threebody.py` test `sp.simplify((f12+f13+f23)-(f23+f13+f12)) == 0`, which is commutative addition in $\mathbb{C}$ rather than a nontrivial clustering constraint (replaced by our three-level hierarchy analysis in Check 4 of `audit_good_twobody.py`).

5 Independent Re-Derivation of K_λ and Frame Analysis (Point 7)

5.1 Direct Deformed-Hamiltonian Derivation at Fixed Physical Momenta

Consider the four-parameter quartic-dispersion Hamiltonian family

$$H_{a,b,d,g} = \int_{-\infty}^{\infty} \mathrm{d}x\, \left(\frac{a}{2}\pi^2 + \frac{b}{2}u^2 - gu^3 + \frac{d}{2}u_x^2\right), \qquad u = \varphi_x, \quad \pi = \varphi_t, \tag{5}$$

whose canonical equation of motion is $u_{tt} = abu_{xx} - adu_{xxxx} - 3ag(u^2)_{xx}$. Substituting the Hirota dependent-variable transformation $u = \frac{2d}{g}(\ln\tau)_{xx}$ yields the deformed bilinear equation

$$\left(\frac{1}{a}D_t^2 - bD_x^2 + dD_x^4\right)\tau\cdot\tau = 0, \qquad \tilde{\mathcal{P}}(K, W) = \frac{W^2}{a} - bK^2 + dK^4. \tag{6}$$

A one-soliton solution $\tau_1 = 1+\mathrm{e}^{2(K_j x+W_j t)}$ with $K_j = \sqrt{b/d}\sin\alpha_j$ and $W_j = -\sqrt{a}b/\sqrt{d}\,\epsilon_j \sin\alpha_j\cos\alpha_j$ has physical velocity $\tilde{v}_j = -W_j/K_j = \sqrt{ab}\,\epsilon_j\cos\alpha_j$, topological charge $\tilde{M}_j = \int u\,\mathrm{d}x = \frac{4\sqrt{bd}}{g}\sin\alpha_j$, and canonical field momentum

$$P_j = -\int \pi u\,\mathrm{d}x = \frac{2b^{3/2}}{\sqrt{a}g^2}\epsilon_j \sin^3\alpha_j\cos\alpha_j = p_*\,P^{(0)}(\alpha_j,\epsilon_j), \qquad p_* = \frac{b^2\sqrt{d}}{\sqrt{a}g^2}. \tag{7}$$

For two solitons, the Hirota interaction coefficient $\tilde{A}_{12} = -\tilde{\mathcal{P}}(K_1-K_2, W_1-W_2)/\tilde{\mathcal{P}}(K_1+K_2, W_1+W_2)$ has b^2/d cancelling between numerator and denominator, so $\tilde{A}_{12}(\alpha_1,\alpha_2) = A_{12}(\alpha_1,\alpha_2)$ at fixed angles α_j. However, the physical momenta P_j are held fixed while (a,b,d,g) vary, so $\alpha_j(P_j;a,b,d,g)$ must vary according to $P^{(0)}(\alpha_j) = P_j/p_*$. Meanwhile, the classical action scale multiplying the dimensionless reduced action is $s = \frac{db^{3/2}}{\sqrt{a}g^2}$, so the classical two-soliton phase in the Darboux frame at fixed physical momenta (P_1,P_2) is

$$\delta_{\mathrm{cl}}(P_1,P_2;a,b,d,g) = s\,\delta_0\!\left(\frac{P_1}{p_*},\frac{P_2}{p_*}\right). \tag{8}$$

Using $\partial_{P_j}\delta_0 = -\Delta X_j'$ and defining $D = \sum_{j=1}^2 P_j\Delta X_j' = -\sum_{j=1}^2 P_j\partial_{P_j}\delta_0$, any deformation scales as

$$\delta_{\mathrm{cl}}(P_1,P_2) = \delta_0(P_1,P_2) + \left(\frac{\delta s}{s}\delta_0 + \frac{\delta p_*}{p_*}D\right) + O(\mathrm{deformation}^2). \tag{9}$$

For the four local operator perturbations:

- $\Delta H_\lambda = \lambda\hbar\int u^2\,\mathrm{d}x \implies \delta b = 2\lambda\hbar \implies \frac{\delta s}{s} = \frac{3}{2}\delta b = 3\lambda\hbar$ and $\frac{\delta p_*}{p_*} = 2\delta b = 4\lambda\hbar$, giving

$$K_\lambda = 3\delta_0 + 4\sum_{j=1}^2 P_j\Delta X_j' = \frac{2}{3}(v_2-v_1)\ln A_{12} + 3T_{12} + 4(P_1\gamma_1 M_2 - P_2\gamma_2 M_1), \tag{10}$$

  where we used the exact trigonometric identity $\frac{2}{3}\ell_j - \frac{4P_j}{k_j} = \frac{2}{3}\left[-v_j(1-4k_j^2) - 4\cdot\frac{2}{3}v_j k_j^2\cdot\frac{3}{2}\right] = -\frac{2}{3}v_j$.

- $\Delta H_\eta = \eta\hbar\int u_x^2\,\mathrm{d}x \implies \delta d = 2\eta\hbar \implies K_\eta = 2\delta_0 + 2D$.
- $\Delta H_\zeta = \zeta\hbar\int u^3\,\mathrm{d}x \implies \delta g = -\zeta\hbar \implies K_\zeta = 2\delta_0 + D$.
- $\Delta H_\kappa = \kappa\hbar\int \pi^2\,\mathrm{d}x \implies \delta a = 2\kappa\hbar \implies K_\kappa = -\delta_0 - D$.

Furthermore, by First-Order Rayleigh–Schrödinger / WKB perturbation theory at fixed asymptotic momenta, $K_\lambda = -\int_{-\infty}^{\infty}\mathrm{d}t\int_{-\infty}^{\infty}\mathrm{d}x\,(u_{12}^2-u_1^2-u_2^2)_{\text{fixed }P_j}$. Decomposing this into the connected insertion at fixed angles α_j, $\mathcal{I}_{u^2}^{\mathrm{conn}} = \int\mathrm{d}t\,\mathrm{d}x\,(u_{12}^2-u_1^2-u_2^2) = \frac{4}{3}(v_2-v_1)\ln A_{12} + 2T_{12} = 3\delta_0 + 2D$, plus the classical trajectory shift from holding P_j fixed ($+2D$), we recover $K_\lambda = (3\delta_0+2D)+2D = 3\delta_0+4D$.

At the benchmark point $(\alpha_1,\alpha_2) = (0.3, 0.6)$, our independent 80-digit recomputation gives:

$$\begin{aligned} K_\lambda^{++}(0.3,0.6) &= +0.05008140486841890958182890110409717393603491445172264526\ldots \neq 0, \\ K_\lambda^{+-}(0.3,0.6) &= -0.22517393868334175848239343385708785721495736988172617955\ldots \neq 0, \end{aligned} \tag{11}$$

confirming the values reported in `good_twobody.tex` (§5) and `good_twobody_counterterms.json` to all 55 stored digits.

5.2 Is K_λ Really Antisymmetric, or Is a Symmetric Part Dropped by a Frame Choice?

This question from Point 7 of the audit brief has a sharp two-part answer proved in Check 1c of `audit_good_twobody.py`:

1. **Within the chamber-reversing Darboux frame, no symmetric term is dropped in forming** $3\delta_0 + 4D$: Once $\delta_0(1,2)$ is defined by Eq. (8) of `good_twobody.tex`,

$$\delta_0(1,2) = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12} + \frac{2}{3}M_1M_2(v_1 - v_2), \tag{12}$$

it satisfies $\delta_0(1,2) + \delta_0(2,1) = 0$ identically under full particle swap $(1 \leftrightarrow 2) : (\alpha_1, \epsilon_1) \leftrightarrow (\alpha_2, \epsilon_2)$ because $A_{12} = A_{21}$. Differentiating $\delta_0(1,2) + \delta_0(2,1) = 0$ with respect to P_1 gives $\Delta X_1'(1,2) = -\Delta X_2'(2,1)$, hence $D(1,2) + D(2,1) = 0$ and $K_\lambda(1,2) + K_\lambda(2,1) = 0$ identically.

2. **However,** $\delta_0(1,2)$ **is itself antisymmetric ONLY because a symmetric boundary coboundary was subtracted to define the chamber-reversing Darboux coordinates**: In the raw Hirota tau function $\tau_2 = 1 + f_1 + f_2 + A_{12}f_1f_2$, the topological center-of-mass shift attaches to whichever soliton lies to the right in a given asymptotic chamber ($x \to -\infty$ vs. $x \to +\infty$). As shown in Eq. (4) of `good_twobody.tex` and audited in `phase2/audit_frame_dictionary.tex`, passing from raw Hirota positions X_j to the symmetric Darboux coordinates $X_j' = X_j + \gamma_j \sum_{l \text{ left of } j} M_l$ subtracts the **symmetric boundary coboundary**

$$S_{\rm cb}(1,2) = \frac{1}{2}(\Pi_1 M_2 + \Pi_2 M_1) = -M_1M_2\frac{v_1 + v_2}{2}, \qquad S_{\rm cb}(1,2) = +S_{\rm cb}(2,1), \tag{13}$$

from the unreduced incoming-chamber action phase $\delta_0^{\rm raw}(1,2) = \delta_0(1,2) + S_{\rm cb}(1,2)$. If one works instead in an incoming-ordered (fixed-chamber) coordinate frame that retains $S_{\rm cb}(1,2)$, the fixed-momentum λ-counterterm response acquires a non-zero **symmetric** part:

$$K_\lambda^{\rm sym}(1,2) = \left(3 - 4\sum_{j=1}^{2} P_j\frac{\partial}{\partial P_j}\right) S_{\rm cb}(1,2) = \frac{1}{2}\Big[(3\Pi_1 - 4P_1\gamma_1)\,M_2 + (3\Pi_2 - 4P_2\gamma_2)\,M_1 - 4\Big(\frac{\Pi_1P_2}{\partial_{\alpha_2}P_2}\cos\alpha_2 + \tag{14}$$

At $(\alpha_1, \alpha_2) = (0.3, 0.6)$ in the overtaking channel $(++)$, this frame-dependent symmetric coboundary response is non-zero:

$$K_\lambda^{\rm sym}(0.3, 0.6; ++) = +0.2711917271184108\ldots \neq 0. \tag{15}$$

Thus K_λ is purely antisymmetric *in the chamber-reversing classical Darboux frame* (where the asymptotic out-state reverses chamber ordering relative to the in-state so that $S_{\rm cb}$ cancels), whereas in a fixed-chamber coordinate frame a non-zero symmetric boundary coboundary part is present. We qualify Claim C4 to make this coordinate-frame mechanism explicit.

6 Three-Body Clustering: Reconciling Continuum and Toda Audits (Point 8)

Point 8 of the audit brief asks us to reconcile two statements:

- `phase2/good_twobody.tex` (§6) proves three-soliton phonon transmission factorisation $T_{123}(q) = T_1(q)T_2(q)T_3(q)$ and shows that scalar three-body factorisation $S_3 = S_{12}S_{13}S_{23}$ leaves an antisymmetric pair addition such as $F_{12} = c(P_1M_2 - P_2M_1)$ unfixed.
- `phase2/audit_frame_dictionary.tex` (Claim 5 and Theorem 1, pp. 2–5) proved on the **Toda** side that spatial cluster separability (a non-vanishing 3×3 determinant minor test) together with the inhomogeneous Krein–Wiener–Hopf PDE uniquely fixes the antisymmetric two-body phase $\widehat{\Phi}_q(u_1, u_2)$ and removes the candidate deformation $A_\lambda(y_1, y_2)$.

In Check 4 of `audit_good_twobody.py`, we performed an exact symbolic comparison of both candidate functions under all three constraints. The reconciliation is governed by a strict **three-level hierarchy of three-body constraints**:

Constraint Level	**Mathematical Condition**	**Infinite-rank $A_\lambda(y_1, y_2)$ (`frame_dictionary`)**	**Rank-2 $F_{12} = c(P_1M_2 - P_2M_1)$ (`good_twobody`)**
Level 1: Scalar 3-body phase additivity & phonon transmission $T_{123} = T_1T_2T_3$	$\delta_3 = \delta_{12}+\delta_{13}+\delta_{23}$ commutes for any pair function $F_{ij} = -F_{ji}$	**Passes** (leaves A_λ unfixed)	**Passes** (leaves F_{12} unfixed)
Level 2: Spatial cluster separability of Darboux frame shifts (3×3 minor test)	$\partial_{P_j}\Phi(1,2)$ depends only on soliton 2's additive charges $\Longrightarrow$ $\det_{1\le a,b\le 3}\Phi(y_a, z_b) = 0$	**Fails**: $\det_{3\times 3} A_\lambda = -\frac{18232351733}{8133427676989440} \neq 0$ (excluded at Level 2)	**Passes**: $\det_{3\times 3} F_{12} = 0$ identically (Schmidt rank 2)
Level 3: Microscopic Toda Krein–Wiener–Hopf PDE (finite-density sea + dressing)	$(\partial_{u_1} + \partial_{u_2})\widehat{\Phi}_q(u_1, u_2) = 2\psi_-(u_1)[\widetilde{\Psi}_-(u_2) - \widetilde{\Psi}_-(0)]$ (Schmidt rank 1)	**Fails**: $\det_{2\times 2}(\partial_1 + \partial_2)A_\lambda \neq 0$ (excluded)	**Fails**: $\det_{2\times 2}(\partial_1 + \partial_2)F_{12} = \frac{27\sqrt{10}}{80} \neq 0$ (excluded at Level 3)

Table 3: Reconciliation of the continuum (`good_twobody.tex`) and Toda (`audit_frame_dictionary.tex`) three-body clustering tests across the three constraint levels.

6.1 Detailed Breakdown of the Three Levels

1. **Is the package's clustering test in `good_twobody_threebody.py` as strong as the Toda test? No.** The continuum package tests *Level 1*: (i) exact four-wave Hirota bilinear factorisation (all $3^4 = 81$ coefficients vanish, proving $T_{123}(q) = T_1(q)T_2(q)T_3(q)$ so the continuum mode integral J is pairwise additive without a genuine three-body phonon density), and (ii) scalar phase additivity $\delta_3 = \delta_{12} + \delta_{13} + \delta_{23}$. Because scalar addition is commutative and associative (indeed, lines 94–95 of `good_twobody_threebody.py` literally check `(f12+f13+f23)-(f23+f13+f12) == 0`), *any* pairwise antisymmetric function $F_{ij} = -F_{ji}$ passes Level 1.

2. **Level 2 (Spatial cluster decomposition of Darboux frame generators)**: In `audit_frame_dictiona` (pp. 3–4), a stronger requirement is imposed: if Φ_{ij} is generated by a local canonical transformation to separated asymptotic Darboux coordinates (X_i', P_i), then when soliton 3 is displaced to spatial infinity, the shift $\Delta X_1' = -\partial_{P_1}\Phi_{12}$ induced by soliton 2 can depend only on a finite set of additive one-body charges (M_2, P_2) of soliton 2. Consequently, any Darboux-generated pair phase must have **Schmidt rank** ≤ 2, so all 3×3 minors $\det_{1\le a,b\le 3}\Phi(y_a, z_b)$ must vanish.

   - The `frame_dictionary` candidate $A_\lambda(y_1, y_2) = \frac{4}{9}(y_1 - y_2)(y_1y_2)^{3/2}\ln\frac{\sqrt{y_1}+\sqrt{y_2}}{|\sqrt{y_1}-\sqrt{y_2}|}$ has infinite Schmidt rank due to the non-separable logarithm; at $y = (1, 2, 3)$ and $z = (4, 5, 6)$, we verified in `audit_good_twobody.py` that $\det_{1\le a,b\le 3}[A_\lambda(y_a, z_b)/\frac{4}{9}] = -\frac{18232351733}{8133427676989440} \neq 0$. Thus Level 2 alone rules out $A_\lambda(y_1, y_2)$.
   - By contrast, the `good_twobody` candidate $F_{12} = c(P_1M_2 - P_2M_1)$ is *explicitly* a rank-2 antisymmetric bilinear of the one-body charges (P_j, M_j), shifting $\Delta X_1' \mapsto \Delta X_1' - cM_2$ by the additive topological charge of soliton 2! Its 3×3 determinant $\det_{1\le a,b\le 3} F_{12}(\alpha_a, \alpha_b)$ vanishes identically for all angles $(\alpha_1, \alpha_2, \alpha_3)$ and $(\beta_1, \beta_2, \beta_3)$ (proved symbolically in `audit_good_twobody.py`). Therefore, **even the Level 2 Darboux 3×3 determinant test does NOT exclude** $F_{12} = c(P_1M_2 - P_2M_1)$.

3. **Level 3 (Which Toda ingredient is absent in the continuum?)**: On the Toda side, solitons are holes/particles in a **finite-density Fermi sea** $[-q, q]$ governed by the linear Bethe-ansatz / Krein integral operator $\mathcal{A} = I - \mathcal{K}$ with step-function source $(\mathcal{A}F_p)(x) =$

$\Theta(p-x)$. Because $\partial_p\Theta(p-x) = -\partial_x\Theta(p-x) = \delta(p-x)$, integration by parts on the Wiener–Hopf domain $[-q,q]$ yields the inhomogeneous **Krein PDE** (`audit_frame_dictionary.tex`, Theorem 1):

$$\left(\frac{\partial}{\partial u_1} + \frac{\partial}{\partial u_2}\right)\widehat{\Phi}_q(u_1,u_2) = 2\psi_-(u_1)\Big[\widetilde{\Psi}_-(u_2) - \widetilde{\Psi}_-(0)\Big] \qquad (u_1 > u_2 > 0), \tag{16}$$

whose right-hand side is a **uniquely determined rank-1 product** of one-body edge resolvents ψ_- and $\widetilde{\Psi}_-$, with boundary condition $\widehat{\Phi}_q(u,u) = 0$. Any homogeneous additive deformation $\Delta\Phi(u_1,u_2)$ at fixed one-body resolvents must satisfy $(\partial_{u_1} + \partial_{u_2})\Delta\Phi = 0$ and $\Delta\Phi(u,u) = 0$, which forces $\Delta\Phi \equiv 0$. In particular, for the scaling-limit bilinear $F_{\mathrm{rank2}}(y_1,y_2) = y_1^{3/2}y_2^{1/2} - y_2^{3/2}y_1^{1/2} \propto P_1M_2 - P_2M_1$, we proved in Check 4c of `audit_good_twobody.py` that

$$\left(\frac{\partial}{\partial y_1} + \frac{\partial}{\partial y_2}\right)\left(y_1^{3/2}y_2^{1/2} - y_2^{3/2}y_1^{1/2}\right) = \frac{y_1^2 - y_2^2}{2\sqrt{y_1y_2}}, \tag{17}$$

which is non-zero and has **Schmidt rank 2** (its 2×2 minor at $y = (1,2), z = (4,5)$ is $\frac{27\sqrt{10}}{80} \neq 0$), so it is excluded on the Toda side by the inhomogeneous Krein PDE (16).

Why this ingredient is absent in the continuum calculation: In the continuum background-field expansion around the zero-density vacuum $u = 0$, there is no finite-density Fermi sea $[-q,q]$ and no linear Bethe-ansatz dressing equation expressing the two-soliton phase as a bilinear form of $(I-\mathcal{K})^{-1}$ on a shared interval; instead, the two-soliton one-loop shift is governed by the spacetime functional determinant of $\mathcal{D}_2 = \partial_t^2 - \partial_x^2 + \partial_x^4 + 6\partial_x(u_{12}\partial_x\cdot)$ together with its four-dimensional zero-mode projection and collective-radiation Schur complement $\det(A - BC^{-1}B^{\mathsf{T}})$, which remain uncomputed (`null`). Moreover, as noted in `audit_frame_dictionary.tex` (line 454), even on the Toda side the identification between the microscopic Krein phase $\widehat{\Phi}_q(u_1,u_2)$ and a specific continuum canonical collective-coordinate convention still leaves a rank-2 endpoint-response dictionary open until the continuum boundary-value problem is solved.

7 Audit of the Grid JSON and `null` Entries (Point 9)

In Check 5 of `audit_good_twobody.py`, we audited the full structure and numerical accuracy of `good_twobody_grid.json`, `good_twobody_threebody.json`, and `good_twobody_resonance.json`:

- **Row counts and channel partition**: `good_twobody_grid.json` contains exactly 270 rows: 90 in $(++)$ ($1 \le i_a \neq i_b \le 10$, $\alpha_j = i_j/10$), 90 in $(--)$ ($1 \le i_a \neq i_b \le 10$), 45 in $(+-)$ ($i_a + i_b \le 10$, strictly below the head-on Y-resonance $\alpha_1 + \alpha_2 = \pi/3 \approx 1.04719755$), and 45 in $(-+)$ ($i_a + i_b \le 10$).
- **Honest `null` entries**: Every one of the 270 rows in `good_twobody_grid.json` has `physical_symmetric: null` and `physical_antisymmetric: null`, accompanied by `physical_status: "uncomputed boundary determinant and state map"` and `unknown_additive_antisymmetric: "B12 (mode convention); not assumed zero"`. Likewise, `good_twobody_threebody.json` records `full_threebody_determinant: null` and `good_twobody_resonance.json` records `quantum_phase: null` for all 6 rows of `overtaking_heavy_Y` (where $\alpha_2 > \pi/3$ lies on the spectrally unstable heavy branch). These `null` entries are completely honest: the projected two- and three-soliton functional determinants, collective-radiation Schur complements, and incoming/outgoing state-normalisation maps have not been computed.
- **Independent 80-digit spot checks**: We recomputed 10 representative rows spanning all four orientation sectors $(++, --, +-, -+)$ and near-resonance head-on kinematics $((\alpha_1,\alpha_2) = (0.1,0.9), (0.2,0.8), (0.3,0.7), (0.4,0.6), (0.5,0.5))$ from scratch at 80 decimal digits using our independent primitives. Every numeric field (`A`, `logA`, `I`, `I_symmetric`, `J_antisymmetric`,

`R_mode`, `R_canonical`, `conditional_mode_representative`, `conditional_canonical_representative`, `lambda_coefficient`, `eta_coefficient`, `zeta_coefficient`, `kappa_coefficient`, `tadpole_coefficien` agreed with the cached JSON to $\le 3.81 \times 10^{-54}$, and all 270 rows satisfied exact particle-exchange antisymmetry $f(1,2) + f(2,1) = 0$ to $< 10^{-65}$.

8 Near-Resonance Asymptotics as $A_{12} \to \infty$ or 0^+ (Point 10)

In Check 6 of `audit_good_twobody.py`, we audited all four near-resonance regimes in §7 of `good_twobody.tex` and `good_twobody_resonance.json` for consistency with the semiclassical phase singularity as $A_{12} \to \infty$ or $A_{12} \to 0^+$:

8.1 Head-On Y-Resonance ($\alpha_1 + \alpha_2 = \pi/3 - h$, $h \to 0^+$, $A_{12} \to +\infty$)

In the head-on channel $\epsilon_1 = +1, \epsilon_2 = -1$, the denominator of A_{12} vanishes linearly at $\alpha_1 + \alpha_2 = \pi/3$: setting $\alpha_2 = \alpha_2^* - h$ with $\alpha_2^* = \pi/3 - \alpha_1$, we have $A_{12}(h) = A_{-1}/h + O(1)$ and $\ln A_{12}(h) = -\ln h + \ln A_{-1} + O(h) \to +\infty$.

- **Why δ_0 remains finite while its momentum derivatives diverge**: At $\alpha_1 + \alpha_2^* = \pi/3$, $\ell_1 - \ell_2^* = -\cos 3\alpha_1 - \cos(3(\pi/3 - \alpha_1)) = -\cos 3\alpha_1 - \cos(\pi - 3\alpha_1) = 0$, so $\ell_1 - \ell_2 = 3\sin(3\alpha_1)\, h + O(h^2)$. Consequently, the logarithmic term in $\delta_0 = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12} + T_{12}$ vanishes as $-\frac{2}{3}\sin(3\alpha_1)\, h \ln h \to 0$, and $\delta_0 \to T_{12}^*$ is finite with a branch-point derivative singularity $\partial_{\alpha_j}\delta_0 \sim \ln h$.

- **Logarithmic divergence of $\Delta X_j'$, $R[p]$, and K_λ**: Because $\partial_{P_j}\delta_0 = -\Delta X_j'$ differentiates across the vanishing prefactor $\ell_1 - \ell_2 = O(h)$, the Darboux shifts $\Delta X_1' \sim -\frac{\ln A_{12}}{k_1}$, $\Delta X_2' \sim +\frac{\ln A_{12}}{k_2}$, the recoil $R[p] \sim C_p(\alpha_1, \alpha_2^*)\ln A_{12} + B_p(\alpha_1, \alpha_2^*) - J_*$, and the counterterm response $K_\lambda \sim C_\lambda(\alpha_1, \alpha_2^*)\ln A_{12} + B_\lambda(\alpha_1, \alpha_2^*)$ all diverge logarithmically as $\ln A_{12} \sim -\ln h$. Meanwhile, the continuum phonon integral $J_{12}(\alpha_1, \alpha_2)$ depends only on the one-soliton factors $T_1(q)$ and $T_2(q)$ and is completely smooth across $\alpha_1 + \alpha_2 = \pi/3$, converging to $J_*(\alpha_1, \alpha_2^*)$.

- **Qualification on the $o(1)$ remainder in Eqs. (22)–(25)**: In `good_twobody_grid.py:89-95`, the subtracted quantities are defined by subtracting the *endpoint* slope $C(\alpha_1, \alpha_2^*)\ln A_{12}(h)$ at $\alpha_2 = \alpha_2^* - h$:

$$\begin{aligned} F_{\rm sub}(h) - B(\alpha_1, \alpha_2^*) &= [C(\alpha_1, \alpha_2^* - h) - C(\alpha_1, \alpha_2^*)]\ln A_{12}(h) + [B(\alpha_1, \alpha_2^* - h) - B(\alpha_1, \alpha_2^*)] \\ &= -h\left[\frac{\partial C}{\partial \alpha_2}(\alpha_1, \alpha_2^*)\, \ln A_{12}(h) + \frac{\partial B}{\partial \alpha_2}(\alpha_1, \alpha_2^*)\right] + O(h^2 \ln h). \end{aligned} \tag{18}$$

  Because $\ln A_{12}(10^{-12}) \approx 31.17$, the leading remainder at $h = 10^{-12}$ is $O(h \ln A_{12}) + O(h) \sim 1.1 \times 10^{-11}$ to 2.0×10^{-11} rather than $O(h) \sim 10^{-12}$. As shown in Table 4, subtracting the exact first-order Taylor correction $-h[(\partial_{\alpha_2} C)\ln A_{12}(h) + \partial_{\alpha_2} B]$ reduces the $h = 10^{-12}$ residual by **twelve orders of magnitude**, to $\le 3.05 \times 10^{-23}$ ($O(h^2 \ln h)$).

α_1	$\lvert F_{\rm mode}^{\rm sub} - B_{p_{\rm m}}\rvert$	After $-h(\partial_{\alpha_2} C)\ln A$	After full $O(h \ln A + h)$	$\lvert K_\lambda^{\rm sub} - B_\lambda\rvert$	After $-h(\partial_{\alpha_2} C_\lambda)\ln A$	After fu
0.2	1.1260×10^{-11}	5.7592×10^{-12}	1.0758×10^{-23}	1.8798×10^{-11}	5.6174×10^{-12}	
0.4	1.6635×10^{-11}	8.0484×10^{-13}	5.9792×10^{-24}	1.4671×10^{-11}	3.9198×10^{-12}	
0.5	2.0328×10^{-11}	1.3833×10^{-12}	1.5254×10^{-23}	1.3240×10^{-11}	3.9471×10^{-12}	

Table 4: Subleading Taylor diagnostics at $h = 10^{-12}$ on the head-on Y-resonance line $\alpha_2 = \pi/3 - \alpha_1 - h$ from Check 6 of `audit_good_twobody.py`. At $\alpha_1 = 0.2$ ($\alpha_2^* \approx 0.8472$), $p_{\rm m}(\alpha_2)/k_2(\alpha_2) = \cot\alpha_2\, E_{\rm m}(\alpha_2)$ is near its stationary point, so the $O(h \ln A_{12})$ and $O(h)$ terms have comparable magnitude; subtracting both reduces all residuals to $O(h^2 \ln h) \le 3.05 \times 10^{-23}$.

8.2 Overtaking Limits ($A_{12} \to 0^+$) and the False $|\alpha_1 - \alpha_2| = \pi/3$ Line

1. **Light equal-velocity overtaking threshold ($\alpha_1 \in (0, \pi/3)$, $\alpha_2 = \alpha_1 + h$, $h \to 0^+$)**: Here $A_{12}(h) = c_0(\alpha_1)h^2 + O(h^3) \to 0^+$, so $\ln A_{12}(h) \sim 2 \ln h \to -\infty$ and the individual Darboux shifts $\Delta X_j' \sim \mp \frac{2}{k_1} \ln h$ diverge. However, because the two solitons approach identical parameters ($\alpha_2 \to \alpha_1$, $\epsilon_2 = \epsilon_1$), the prefactors in $R[p] = (p_2/k_2 - p_1/k_1) \ln A_{12} + O(h)$ and $K_\lambda = \frac{2}{3}(v_2 - v_1) \ln A_{12} + O(h)$ vanish linearly in h:

$$K_\lambda(\alpha_1, \alpha_1{+}h; {+}{+}) = -\frac{4}{3} \sin \alpha_1 \, h \ln h {+} O(h) \to 0, \qquad R[p](\alpha_1, \alpha_1{+}h; {+}{+}) = O(h \ln h) \to 0, \qquad J_{12} = O(h \tag{19}$$

   At $\alpha_1 = 0.4, h = 10^{-12}$, $K_\lambda = +1.434298 \times 10^{-11}$, matching $-\frac{4}{3} \sin(0.4)\, h \ln h = +1.434663 \times 10^{-11}$ to $3.65 \times 10^{-15} = O(h)$.

2. **Heavy overtaking Y-line ($\alpha_1 + \alpha_2 = 2\pi/3 - h$, $h \to 0^+$)**: Because $\alpha_1 + \alpha_2 = 2\pi/3$ cannot be reached with both $\alpha_j \in (0, \pi/3)$, at least one soliton must lie on the spectrally unstable heavy branch $\alpha > \pi/3$. At $\alpha_1 = 0.9$, $\alpha_2^* = 2\pi/3 - 0.9 \approx 1.194395 > \pi/3$, $A_{12}(h) \propto h \to 0^+$ while $v_2^* - v_1 = \cos \alpha_2^* - \cos \alpha_1 \approx -0.254266 \neq 0$, so $K_\lambda(h) \sim \frac{2}{3}(v_2^* - v_1) \ln h \to +\infty$ diverges logarithmically (increasing by $-\frac{2}{3}(v_2^* - v_1) \ln 100 \approx +0.779913$ per factor of 100 in h). Because the heavy branch has tachyonic internal modes, `good_twobody_resonance.json` rightly sets `quantum_phase: null`.

3. **Refutation of the false overtaking resonance $|\alpha_1 - \alpha_2| = \pi/3$**: At $(\alpha_1, \alpha_2) = (0.2, 0.2 + \pi/3)$ with $\epsilon_1 = \epsilon_2 = +1$, $A_{12} = 0.35546547984906165\ldots$, which is strictly positive and finite ($0 < A_{12} < 1$). The numerator of A_{12} factors via $\cos 3\alpha_1 - \cos 3\alpha_2 = 2 \sin \frac{3(\alpha_2 - \alpha_1)}{2} \sin \frac{3(\alpha_1 + \alpha_2)}{2}$, whose zeroes in the overtaking channel are $\alpha_2 - \alpha_1 = 0$ and $\alpha_1 + \alpha_2 = 2\pi/3$, *not* $|\alpha_1 - \alpha_2| = \pi/3$.

9 Failures Recorded

In compliance with `AGENTS.md` ("Report failures honestly. List failed checks, including ones that were later fixed, in the write-up"), we record both the historical failures documented by the package and all seven failures encountered and fixed during the development of `phase2/scripts/audit_good_twobody.py`.

9.1 Historical Failures Documented in `good_twobody.tex`

1. **Legacy fixed-momentum counterterm vanishing claim**: Earlier exploratory Phase 2 notes claimed that $\Delta H = \lambda \hbar \int u^2 \, \mathrm{d}x$ produced zero shift of the two-soliton phase at fixed physical momenta. Direct scaling and spacetime integration refute this: $K_\lambda = 3\delta_0 + 4 \sum_j P_j \Delta X_j' \neq 0$.

2. **Legacy overtaking resonance at $|\alpha_1 - \alpha_2| = \pi/3$**: Earlier Phase 2 notes placed an overtaking resonance at $|\alpha_1 - \alpha_2| = \pi/3$. Direct evaluation gives $A_{12}(0.2, 0.2 + \pi/3; {+}{+}) \approx 0.355465 \neq 0, \infty$; the actual zeros/poles of A_{12} are $\alpha_1 + \alpha_2 = \pi/3$ (head-on Y-resonance), $\alpha_1 = \alpha_2$ (overtaking threshold), and $\alpha_1 + \alpha_2 = 2\pi/3$ (heavy overtaking Y-line).

3. **Unjustified setting of $B_{12} = 0$ and physical symmetric phase to zero**: Earlier notes set the unknown boundary-projection phase B_{12} to zero without computing the projected two-soliton determinant or collective-radiation Schur complement. The audited package replaces those entries with explicit `null` values.

9.2 Failures Encountered and Fixed During This Independent Audit

1. **Run 1 of `audit_good_twobody.py` (`SyntaxError` at line 317)**: A typographic space inside the identifier `ddelta0_da1_ minus_log` caused a Python syntax error. Fixed by removing the space (`ddelta0_da1_minus_log`).

2. **Run 2 of `audit_good_twobody.py` (`AssertionError` and `ValueError` at lines 270–271 in Check 1a)**: `sp.trigsimp((sp.diff(E1_true, a1) - v1 * sp.diff(P1, a1)).subs(e1**2, 1))` did not simplify to 0 because `v1 * sp.diff(P1, a1)` was not expanded before substituting `e1**2: 1`, and `sp.diff(Pi1, k1)` raised `ValueError: Can't calculate derivative wrt sin(a1)`. Fixed by calling `sp.expand` before `.subs(e1**2, 1)` and computing $\partial_{k_1}\Pi_1 = (\partial_{\alpha_1}\Pi_1)/\cos\alpha_1$.

3. **Run 3 of `audit_good_twobody.py` (`AssertionError` at lines 306 and 325 in Check 1b)**: With symbolic direction signs ϵ_1, ϵ_2, `.subs({e1**2: 1, e2**2: 1})` left odd powers ϵ_j^3 unreduced, and `sp.trigsimp` did not canonicalise rational trigonometric expressions mixing $\sin 3\alpha_j$ and $\sqrt{3}\sin\alpha_j$. Fixed by looping explicitly over the four sign pairs $(\epsilon_1, \epsilon_2) \in \{(1,1),(1,-1),(-1,1),(-1,-1)\}$ with symbolic (α_1, α_2) and applying `.rewrite(sp.exp)` followed by `sp.expand`.

4. **Run 4 of `audit_good_twobody.py` (`AssertionError` at line 433 in Check 2a)**: Our initial implementation of `deformed_twobody_phase_fixed_P` wrote the topological term T_{12} with prefactor $\frac{1}{3}(\Pi_2 M_1 - \Pi_1 M_2)$ instead of $\frac{2}{3}(\Pi_2 M_1 - \Pi_1 M_2)$, producing a 0.1065 discrepancy against K_λ. Fixing the prefactor to $\frac{2}{3}$ reduced the deformed-PDE Richardson error across all four counterterms $(\lambda, \eta, \zeta, \kappa)$ to 3.05×10^{-54}.

5. **Run 5 of `audit_good_twobody.py` (`AssertionError` at line 499 in Check 2c)**: In the symbolic expansion of the one-loop ring $\frac{1}{2}\ln(1 - \frac{6Uq^2}{\omega_{\mathrm{E}}^2 + M^2})$, we initially wrote the n-th Taylor coefficient with alternating sign $(-1)^n/n$ instead of $-1/(2n)$. Fixing the Taylor coefficient to $-1/(2n)$ gave exact symbolic agreement for $n = 1, 2, 3, 4$.

6. **Run 6 of `audit_good_twobody.py` (`AssertionError` at line 812 in Check 6a)**: We initially asserted that subtracting the $O(h \ln A_{12})$ term alone would reduce both the mode and λ remainders at $h = 10^{-12}$ by $> 5\times$ at all $\alpha_1 \in \{0.2, 0.4, 0.5\}$. At $\alpha_1 = 0.2$ ($\alpha_2^* = \pi/3 - 0.2 \approx 0.8472$), $p_{\mathrm{m}}(\alpha_2)/k_2(\alpha_2) = \cot\alpha_2\, E_{\mathrm{m}}(\alpha_2)$ is near a stationary point, so the $O(h \ln A_{12})$ and $O(h)$ terms have comparable magnitude (Wait: subtracting both the $O(h \ln A_{12})$ and $O(h)$ Taylor terms reduces the residual from 1.13×10^{-11} to 1.08×10^{-23}). Fixed by testing the full first-order Taylor subtraction $O(h \ln A_{12}) + O(h)$ to $< 10^{-21}$.

7. **Run 7 of `audit_good_twobody.py` (`AssertionError` at line 844 in Check 6c)**: On the heavy overtaking Y-line, we initially asserted that the step difference $K_\lambda(10^{-12}) - K_\lambda(10^{-10})$ matched $-\frac{2}{3}(v_2^* - v_1)\ln 100$ to $< 10^{-10}$, overlooking that the subleading correction at $h_1 = 10^{-10}$ is $O(h_1 \ln h_1) \sim 2.3 \times 10^{-9}$. Fixed by setting the step-difference bound to 5×10^{-9} and additionally verifying that $K_\lambda(10^{-12}) - C_\lambda^* \ln A_{12}(10^{-12})$ matches $B_\lambda^{\mathrm{heavy}}(\alpha_1, \alpha_2^*)$ to $< 10^{-9}$.

8. **Initial `pdflatex` passes on `audit_good_twobody.tex`**: Pass 1 failed on `Undefined color 'blue!60!black'` (because `xcolor` was not loaded) and a stray underscore `_` at line 286 (`continuum?)_`). Fixed by using standard color `blue` and removing the stray underscore.

References

[1] R. Hirota, "Exact N-soliton solutions of the wave equation of long waves in shallow-water and in nonlinear lattices," *J. Math. Phys.* **14** (1973) 810–814, doi:10.1063/1.1666400. (Verified entry and classical bilinear formulas against project conventions.)

[2] J. L. Bona and R. L. Sachs, "Global existence of smooth solutions and stability of solitary waves for a generalized Boussinesq equation," *Commun. Math. Phys.* **118** (1988) 15–29, doi:10.1007/BF01218475. (Verified spectral stability threshold $\alpha \in (0, \pi/3)$ for the light branch vs. heavy branch $\alpha > \pi/3$.)

[3] E. Corrigan and C. Zambon, "Integrable defects in the Boussinesq equations," arXiv:2306.03772 [hep-th]. (Full text read in repository; dictionary $u_{\mathrm{CZ}} = -3\varphi$, $\mathcal{L}_{\mathrm{CZ}} = 9\mathcal{L}$ verified.)